# Managing the Additive Manufacturing Supply Chain 252

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**Class Outline**

Supply Chain Management and AM
Digital Thread and Digital Twin
Serialization
Serialization and Additive Manufacturing
Review: Supply Chain Management and Additive Manufacturing
Blockchain
Blockchain and Additive Manufacturing
Review: Serialization and Blockchain

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**Supply Chain Management and AM**

Behind every product that reaches a customer, there is an intricate series of exchanges between many different companies. A supplier of raw materials will transfer them to a company that uses them to produce various parts, for example. Then, the parts may travel to a different company that assembles them into a finished product, yet another company that stores the product, and another company that distributes the product to customers. These companies that exchange information, goods, and services with each other in order to create a specific product and make it available to consumers are a supply chain. Supply chain management involves coordinating and optimizing these various activities, from initial material purchases to inventory production and management to final product distribution.

Additive manufacturing (AM) transforms traditional supply chains and their management. AM technology turns digital three-dimensional models into physical objects by building them up in layers. Raw materials used in AM are often in the form of powders or pellets that can be reclaimed and reused if not used in a build. As a result, AM may allow companies to save on material costs. AM may also minimize storage needs and combine production and assembly into fewer steps. The increased responsiveness to customer needs that AM allows can enable greater agility in the manufacturing process.

Supply chain managers must consider the affects AM may have on their supply chain as well as the total cost of ownership (TCO) of new programs and devices. TCO is the purchase price of an asset plus the costs of operation. By evaluating TCO, a supply chain manager can decide if additive manufacturing programs will prove to be efficient investments over time.

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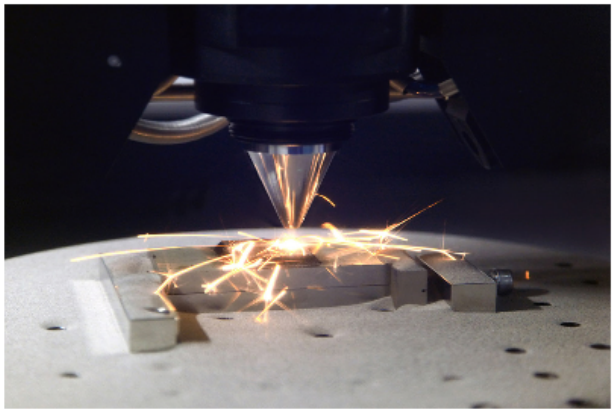

**Figure 1:** AM with Metal

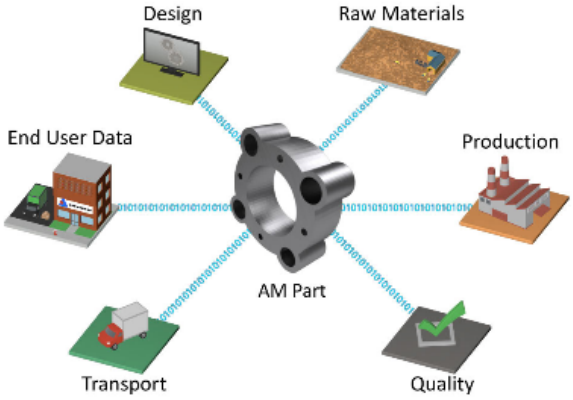


**Figure 2:** The AM Supply Chain

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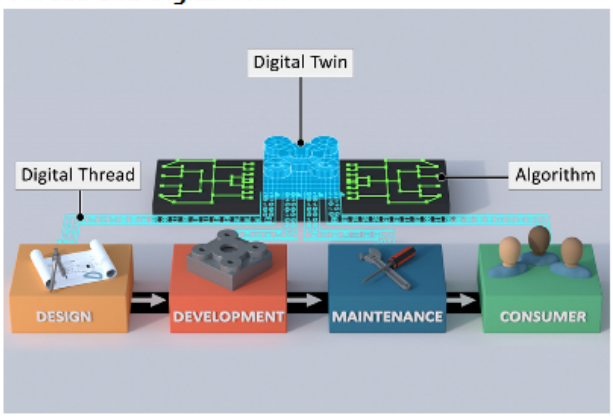
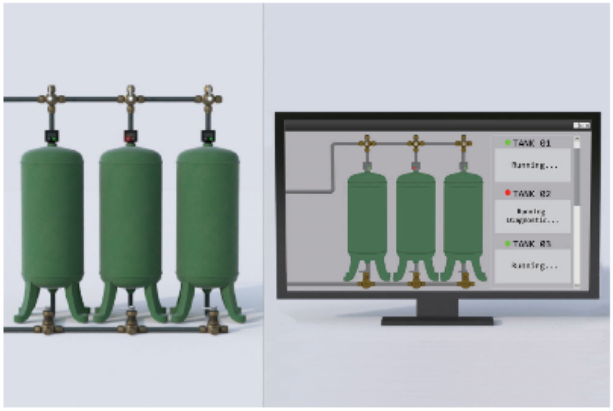
## Digital Thread and Digital Twin




A digital thread is a framework that maintains data on a product from its design to its final use, which produces a comprehensive view of each product's lifecycle. This framework addresses protocols, security, and standards. Digital threads may connect design, development, maintenance, and consumers. Digital threads often work in conjunction with a digital twin, a virtual representation of a physical object.



Supply chains may also have their own digital twins, which act as a virtual model of every segment of the process. Working with digital twins, a digital thread enables manufacturers to quickly adjust to disruptions or inventory problems. Instead of human feedback, manufacturers often depend on computer-based algorithms to manage the constant stream of data. Algorithms can instantly deploy a response to issues along the digital supply chain (DSC). As a result, supply chain management in the digital thread environment significantly increases efficiency and reduces the need for human input.

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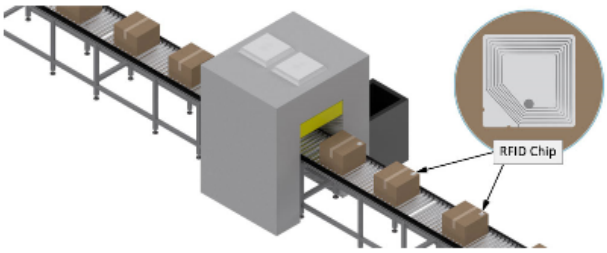
**Lesson:** 3/8

**Serialization**




Serialization is the process of assigning unique, sequential identifiers to each distinct unit of inventory that a facility owns or manufactures. Often, these identifiers take the form of serial numbers, though they may also be non-numerical tags, such as radio frequency identification (RFID) chips.

Serial numbers may be included on individual parts or components, as well as final assembled products. Serialization allows for traceability throughout the supply chain, because manufacturers can use serial numbers or tags to track items back to their original production sites. Manufacturers can read the serialized codes either manually or by machine, and can identify which supply chain entities contributed to, stored, or transported the products. Some codes or identifiers may also include data on the specific parts that may have been used in the product's build. Ultimately, serialization allows supply chain partners to track each product's lifecycle.

The traceability that serialization provides helps to guard against counterfeiting and piracy, which can weaken a product's reputation and hurt business if consumers receive poor-quality fakes. Serialization also allows for quick consumer recall if the manufacturer discovers quality or safety issues with a product or its parts.

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**Serialization and Additive Manufacturing**

Each AM machine has a specific processing framework based on parameters including part materials, part geometry, and build method. Normally, AM processes are optimized for precise part creation. However, these processing parameters can be temporarily altered to create a unique identifying mark within the part material that is virtually impossible to replicate. Unless the part is broken or cut open, this mark can only be viewed using various nondestructive testing (NDT) techniques such as X-rays. After a part is manufactured, it would be scanned to record the unique mark. The manufacturer can then keep a digital copy of that scan, which can be used to compare future scans to confirm the part's authenticity.

Serialization can code and identify both manufactured parts and AM machines themselves. Serializing AM machines allows supply chain entities to track elements such as a build's initial designer. It acts as a traceable link between the part and the conditions of its creation.

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**Lesson:** 5/8

**Review: Supply Chain Management and Additive Manufacturing**

- **Additive manufacturing**: The process of joining or solidifying materials to make an object based on a 3-D computer model.
- **Digital thread**: An integrated view of all the data and information about a part or product throughout its lifecycle.
- **Total cost of ownership**: The purchase price of an asset plus the costs of operation.
- **Serialization**: The process of tagging individual units with unique, sequentially assigned identifiers.

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**Lesson:** 6/8

**Blockchain**
Like serialization, blockchain is a method that increases traceability, accountability, and security against counterfeits and piracy. Unlike serialization, blockchain is entirely digital. Instead of just adding a single, unique identifier to each product or machine, it ties all the identifiers together in a digital chain to make fraud difficult.

Blockchain technology is a decentralized, distributed ledger, which means that it is a record stored on a network of computers rather than on a single device. This helps to prevent records from being stolen or destroyed. Blockchain records the provenance, or place of origin, of a digital asset.

A block in a blockchain essentially acts as a digital, rather than physical, serial number for an item. Blocks act as links in the chain. Every chain includes several blocks, and each block consists of three elements:

- data about the provenance of the asset.
- a number called a nonce. Creating a block generates a random nonce.
- the block header hash, a much larger number that is attached to the nonce. Just as creating a block generates a nonce, a nonce in turn generates the block header hash.

Since a blockchain is a distributed ledger, no single individual or site can single-handedly control the chain. Instead, the chain's information is distributed across a network of nodes. A node is any kind of computer or electronic device that preserves copies of the blockchain and maintains the functioning of the network.

Every action in the ledger can be reviewed, which makes blockchain remarkably transparent. Each contributor to the chain is given an exclusive alphanumeric identification number that records their transactions. Because of this, all alterations to the blockchain are traceable, which builds user confidence in the chain's integrity.

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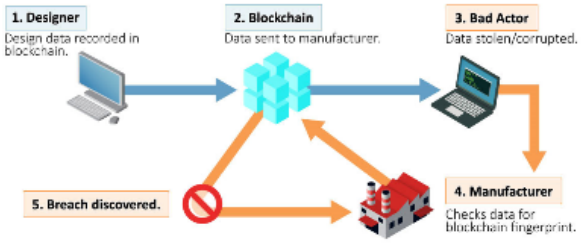
**Lesson:** 7/8

**Blockchain and Additive Manufacturing**




In traditional manufacturing supply chains, the company that creates a part design would also handle manufacturing and then shipping the part. In the AM supply chain, however, the part designers may digitally transmit design files to other organizations that are part of the supply chain. Although most companies who send digital design files are rigorous about cybersecurity, digital transfer always carries risk. Blockchain helps to prevent a design file from being compromised in the event of a cybersecurity breach. Blockchain transactions allow the file's distribution to be authenticated, transported, and recorded. It also enables all members of the blockchain to recognize the origin of the design data.

Engineers can also use blockchain to apply production rules to the design files that specify build parameters, including the make and model of the machine allowed to execute the design and the types of build materials permitted. Because the files are sent directly to the AM machines, manufacturers are only able to access the design files once these specifications are met. Additionally, production rules control the number of parts each manufacturer is licensed to print. This ensures that all parts meet quality standards, and it prevents counterfeits from being made on unauthorized equipment. Finally, the blockchain's distributed ledger maintains records of every action and change throughout a design's lifecycle. This allows designers and manufacturers to authenticate a part's origin, which means that any problems in a finished part can be located and addressed in the production process.

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**Lesson:** 8/8

**Review: Serialization and Blockchain**

1. A block from a blockchain contains data, a nonce, and a hash. **TRUE**
2. Blockchains are maintained on one centralized computer system. **FALSE**
3. Blockchain technology allows for increased transparency. **TRUE**
4. Blockchain technology is less secure than serialization. **FALSE**
5. Engineers can use serialization to apply rules about what machines can execute design files. **FALSE**
6. Blockchain allows errors in end products to be traced to their source. **TRUE**

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**Class Vocabulary**

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| **Additive Manufacturing** | AM. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object. |
| **Agility** | The ability to respond quickly to customer needs and market changes while still controlling costs and quality. Agility in manufacturing may be enabled by additive manufacturing. |
| **Algorithms** | A mathematical process designed to systematically solve a problem. Complex digital algorithms allow machine learning to predict and regulate operations. |
| **AM** | Additive manufacturing. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. AM methods typically build up layers of material to create an object. |
| **Block Header Hash** | A 256-bit number attached to the nonce when a block is created. The hash is an essential element of a block in blockchain technology. |
| **Blockchain** | A growing list of digital records, called blocks, that are linked. Blockchain technology is a decentralized, distributed ledger that records the origin of a digital asset. |
| **Blocks** | One identifying unit within a blockchain. Blocks include data and unique identifying numbers. |
| **Counterfeiting** | Imitating fraudulently. Counterfeiting involves making an exact imitation of something valuable with the intention to deceive or defraud. |
| **Cybersecurity** | Protection against criminal or unauthorized access to computer networks, programs, and data. Cybersecurity has become a major industrial concern as networking and connectivity have increased. |
| **Decentralized** | The allocation of resources, both hardware and software, to each individual workstation or office location. Decentralized computer systems are components of a larger computer network. |
| **Digital Supply Chain** | DSC. A web-based network of computers, companies, and systems that exchange resources to deliver products to customers. A digital supply chain connects suppliers and stakeholders throughout the entire product lifecycle. |
| **Digital Thread** | An integrated view of all the data and information about a part or product throughout its lifecycle. The digital thread connects information from all aspects of a product into one seamless network. |
| **Digital Twin** | A virtual representation of a physical object, such as a part or machine. A digital twin evolves with the object throughout its lifecycle. |
| **Distributed Ledger** | A consensus of replicated, shared, and synchronized digital data geographically spread across multiple sites, countries, or institutions. In a distributed ledger, there is no central administrator or centralized data storage. |
| **Lifecycle** | The entire timeline of something. In manufacturing, lifecycle includes part design, machine setup, production, quality, and end of life. |
| **Nodes** | Any kind of electronic device that maintains copies of the blockchain and keeps the network functioning. Nodes are decentralized and distributed. |
| **Nonce** | A 32-bit whole number randomly generated when a block is created, which then generates a block header hash. A nonce is an essential element of a block in blockchain technology. |
| **Nondestructive Testing** | NDT. A category of inspection processes that evaluate a part's properties and performance using methods that do not damage or permanently alter the part. Nondestructive testing methods include visual testing, liquid penetrant testing, magnetic particle testing, eddy current testing, ultrasonic testing, and radiographic testing. |
| **Piracy** | The unauthorized use or reproduction of another's work. Piracy is the illegal copying, distribution, or use of intellectual property. |
| **Provenance** | The place of origin of something. Provenance of a digital asset is recorded in blockchain. |
| **Radio Frequency Identification** | RFID. A technology that uses electromagnetic fields to automatically identify and track tags attached to |

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| **Radio Frequency Identification** | RFID. A technology that uses electromagnetic fields to automatically identify and track tags attached to objects. RFID tags consist of a tiny radio transponder, radio receiver, and radio transmitter. |
| **Regulations** | A rule or directive made and maintained by an authority. Regulations can be set by governmental agencies and industry organizations. |
| **Regulatory Compliance** | When a business follows state, federal, and international laws and regulations relevant to its operations. Regulatory compliance varies by industry, as well as the countries where a business sells products. |
| **Serial Numbers** | A unique identifier assigned incrementally or sequentially to an item in a series. Serial numbers need not be strictly numerical and may contain letters or other symbols. |
| **Serialization** | The process of tagging individual units of inventory with unique identifiers that are assigned sequentially to each item. Serialization may take the form of serial numbers, RFID chips, or other unique, sequential tags. |
| **Supply Chain** | A network of companies that exchange resources, such as materials and information, to deliver products to customers. Supply chains consist of a company, its suppliers, its distributors, and its customers. |
| **Supply Chain Management** | The process of planning, implementing, and controlling supply chain activities to achieve maximum customer value and sustain competitive advantage. Supply chain management oversees each organization in the supply chain, from development to sourcing to production to delivery. |
| **Total Cost Of Ownership** | TCO. The purchase price of an asset plus the costs of operation. Total cost of ownership helps assess the value of an investment over time. |
| **Traceability** | The ability to verify the history, application, or location of an item using data flows. Digital threads enable traceability throughout the entire production lifecycle. |
| **Virtual** | Existing on or simulated within a computer program or system rather than physically existing. Virtual representations of real-world objects are known as digital twins. |
| **X-Rays** | A type of electromagnetic wave with a high frequency and short wavelength. X-rays are used to view the interior of solid objects during radiographic testing. |

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